

DE Midterm
Hayden Fuller

P1:

a) $V_{T0} = -0.3V$, $\gamma = .3626 V^{1/2}$, $E_{cL} = 2.1798V$, $\lambda = 0.1623 V^{-1}$, $\mu_p = 67.5 \text{ cm}^2/Vs$
i)

V_{T0} :

using plot 1 at $V_{SB}=0$, in saturation the short channel equation is

$$I_{DS} = W \mu_{sat} C_{ox} ((V_{GS}-V_T)^2)/((V_{GS}-V_T)+E_{cL})$$

at the cusp of $I_{DS} = 0$, $V_{GS}-V_T=0$, $V_{GS}=V_T=V_{T0}=-0.3V$

$$V_{T0} = -0.3V$$

γ :

using plot 1 comparing V_{SB} curves, the body effect equation

$$V_T = V_{T0} + \gamma (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$

$$V_T - V_{T0} = \gamma (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$

$$V_{SB}=0V \rightarrow V_T = -0.3V \rightarrow V_T - V_{T0} = 0V$$

$$V_{SB}=-0.4V \rightarrow V_T = -0.37V \rightarrow V_T - V_{T0} = -0.07V$$

$$V_{SB}=-0.8V \rightarrow V_T = -0.43V \rightarrow V_T - V_{T0} = -0.13V$$

since we have desired points (0,0), (0.4, 0.07), (0.8, 0.13) from

$$\Delta V_T(\gamma) = (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$

$$\text{find minimum variance } ((x(\sqrt{.88+.4})-\sqrt{.88})-.07)/.07)^2 + ((x(\sqrt{.88+.8})-\sqrt{.88})-.13)/.13)^2$$

$$\gamma = .3626 V^{1/2}$$

E_{cL} :

using plot 2 boundary between linear and saturation

$$V_{DSsat} = ((V_{GS}-V_T)E_{cL})/((V_{GS}-V_T)+E_{cL})$$

since $V_{SB}=0V$, $V_T=0.3V$

$$V_{GS}=1.2: V_{DS} = -.637V$$

it's much harder to pinpoint the exact corner of the other curves with lower slopes, so I'll stick with this one. I plugged in my final number and got V_{DSsat} 's of .530, .407, and .264, which all seem to about line up with the plot

$$0.637 = ((1.2-0.3)E_{cL})/((1.2-0.3)+E_{cL})$$

$$E_{cL} = 2.1798V$$

λ :

using plot 2 linear region

same as for E_{cL} , I'll just use $V_{GS}=1.2$ since the others are barely measurable

$$V_{DS} = -1$$

$$I_{DS} = 305.5\mu A$$

$$V_{DSsat} = 0.637V$$

$$I_{DSsat} = 288.5\mu A$$

$$\lambda = \text{slope}/I_{DSsat} = ((305.5\mu - 288.5\mu)/(1-.637))/288\mu = 0.1623$$

$$\lambda = 0.1623 V^{-1}$$

ii)

u_p :

using plot 1

$k' = u C_{ox}$

$u = k' / C_{ox}$

$I_{DS} = k' / 2 W/L (V_{GS} - V_T)^2$

$\sqrt{I_{DS}} = \sqrt{k' / 2 W/L} (V_{GS} - V_T)$

$\sqrt{2 I_{DS}} = \sqrt{k' W/L} (V_{GS} - V_T)$

$\text{slope} = \sqrt{k' W/L} = 23 \text{mA}^{1/2} / .7\text{V} = 32.857 \text{mA}^{1/2}/\text{V}$

$k' = L/W \text{slope}^2$

$k' \sim .1 * (32.857 \text{mA}^{1/2}/\text{V})^2$

$k' \sim 0.4 * (32.857 \times 10^{-3} \text{A}^{1/2}/\text{V})^2 = 108 \times 10^{-6} \text{A}/\text{V}^2$

$u = (108 \times 10^{-6} \text{A}/\text{V}^2) / (1.6 \times 10^{-6} \text{C}/\text{Vcm}^2) = (108/1.6) \text{cm}^2/\text{Vs}$

$u_p = 67.5 \text{cm}^2/\text{Vs}$

b)

```
.model PMOS2 PMOS
```

```
+Level = 2
```

```
+vt0 = -0.30001
```

```
+gamma = 0.3626
```

```
+nsub = 3E17
```

```
+lambda = 0.1623
```

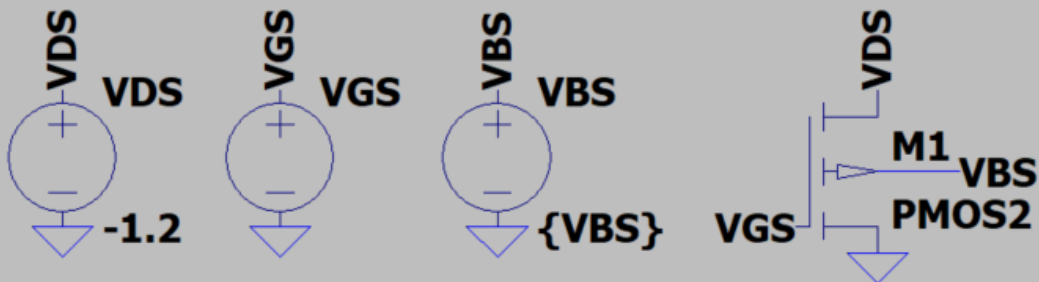
```
+tox = 2.2E-09
```

```
+vmax = 8E4
```

```
+Uo = 67.5
```

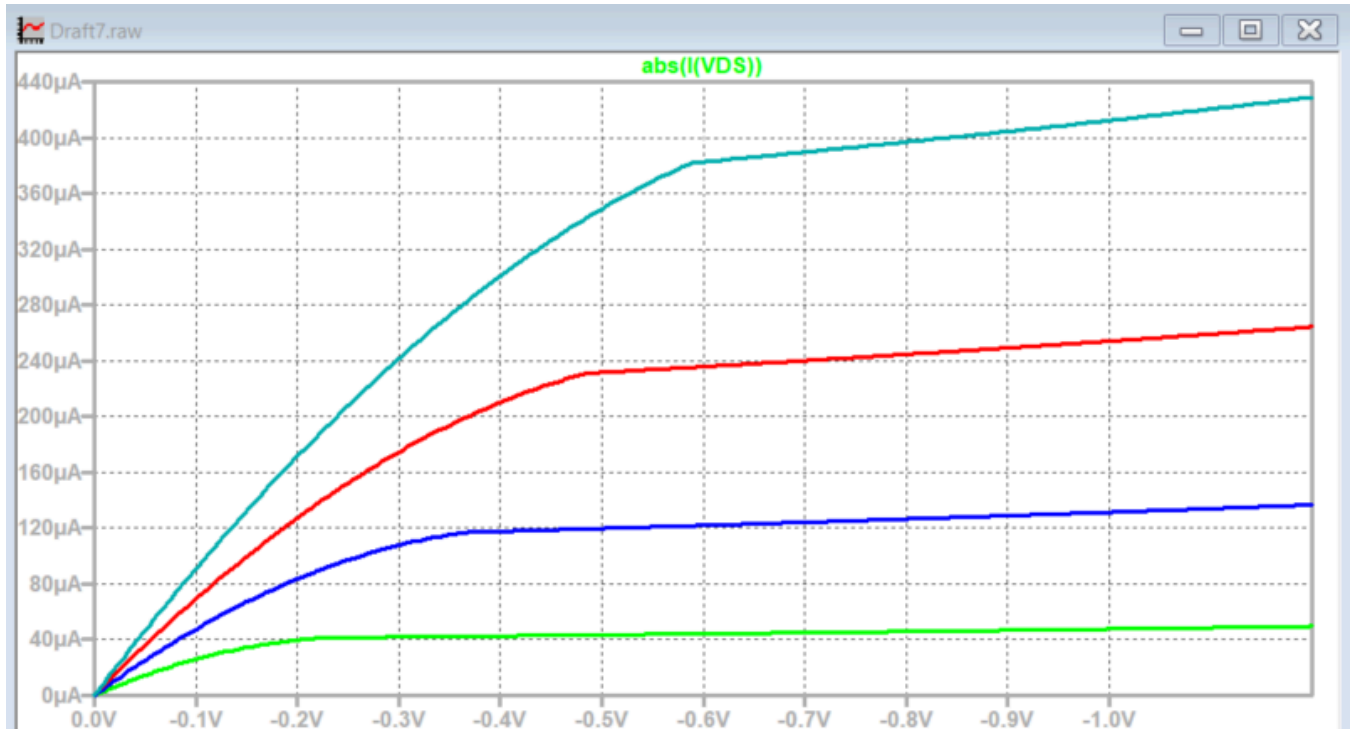


```
.include C:\Users\Hayden\Desktop\DE\Week5\MOS2midterm.txt
```



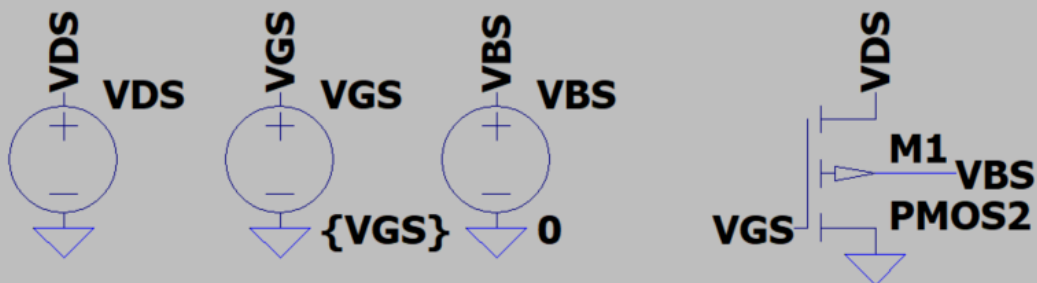
```
.step param VBS list 0 0.4 0.8
```

```
.dc VGS 0 -1.2 0.01
```



this plot is off by a factor of $\sqrt{2}$, not exactly sure how that got in there
 it can be aided by dropping μ_p by $1/\sqrt{2}$, but that messes with the first plot?

.include C:\Users\Hayden\Desktop\DE\Week5\MOS2midterm.txt



.step param VGS list -0.6 -0.8 -1.0 -1.2
.dc VDS 0 -1.2 0.01

P2:

a) 1.85355, 0.9259259, 0.50223

linear enhancement load NMOS inverter

driver current (saturation)

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W_D}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$I_D = \frac{1}{2} * 270 * 1.6 \times 10^{-6} * \frac{W_D}{100} * (1.2 - 0.4)^2 (1 + \lambda * V_{out})$$

$$I_D = \frac{1}{2} * 270 * 1.6 \times 10^{-6} * \frac{W_D}{100} * (1.2 - 0.4)^2 (1 + 0.2 * V_{out})$$

$$I_D = 1.3824 \times 10^{-6} W_D (1 + 0.2 * V_{out})$$

load current (linear)

$$I_L = \mu_n C_{ox} \frac{W_L}{L} ((V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2})$$

$$I_L = 270 * 1.6 \times 10^{-6} * \frac{W_L}{100} (((1.6 - V_{out}) - 0.4)(1.2 - V_{out}) - \frac{(1.2 - V_{out})^2}{2})$$

$$I_L = 4.32 \times 10^{-6} * W_L ((1.2 - V_{out})^2)/2$$

$$I_L = 2.16 \times 10^{-6} * W_L ((1.2 - V_{out})^2)$$

i)

$$V_{out} = 0.1V$$

$$I_D = 1.3824 \times 10^{-6} W_D (1 + 0.2 * 0.1) = I_L = 2.16 \times 10^{-6} * W_L ((1.2 - 0.1)^2)$$

$$1.3824 \times 10^{-6} W_D (1.02) = 2.16 \times 10^{-6} * W_L (1.21)$$

$$W_D/W_L = (2.16 \times 10^{-6} * 1.21) / (1.3824 \times 10^{-6} * 1.02)$$

$$W_D/W_L = 1.85355$$

ii)

$$V_{out} = 0.4V$$

$$I_D = 1.3824 \times 10^{-6} W_D (1 + 0.2 * 0.4) = I_L = 2.16 \times 10^{-6} * W_L ((1.2 - 0.4)^2)$$

$$1.3824 \times 10^{-6} W_D (1.08) = 2.16 \times 10^{-6} * W_L (0.64)$$

$$W_D/W_L = (2.16 \times 10^{-6} * 0.64) / (1.3824 \times 10^{-6} * 1.08)$$

$$W_D/W_L = 0.9259259$$

iii)

$$V_{out} = 0.4V$$

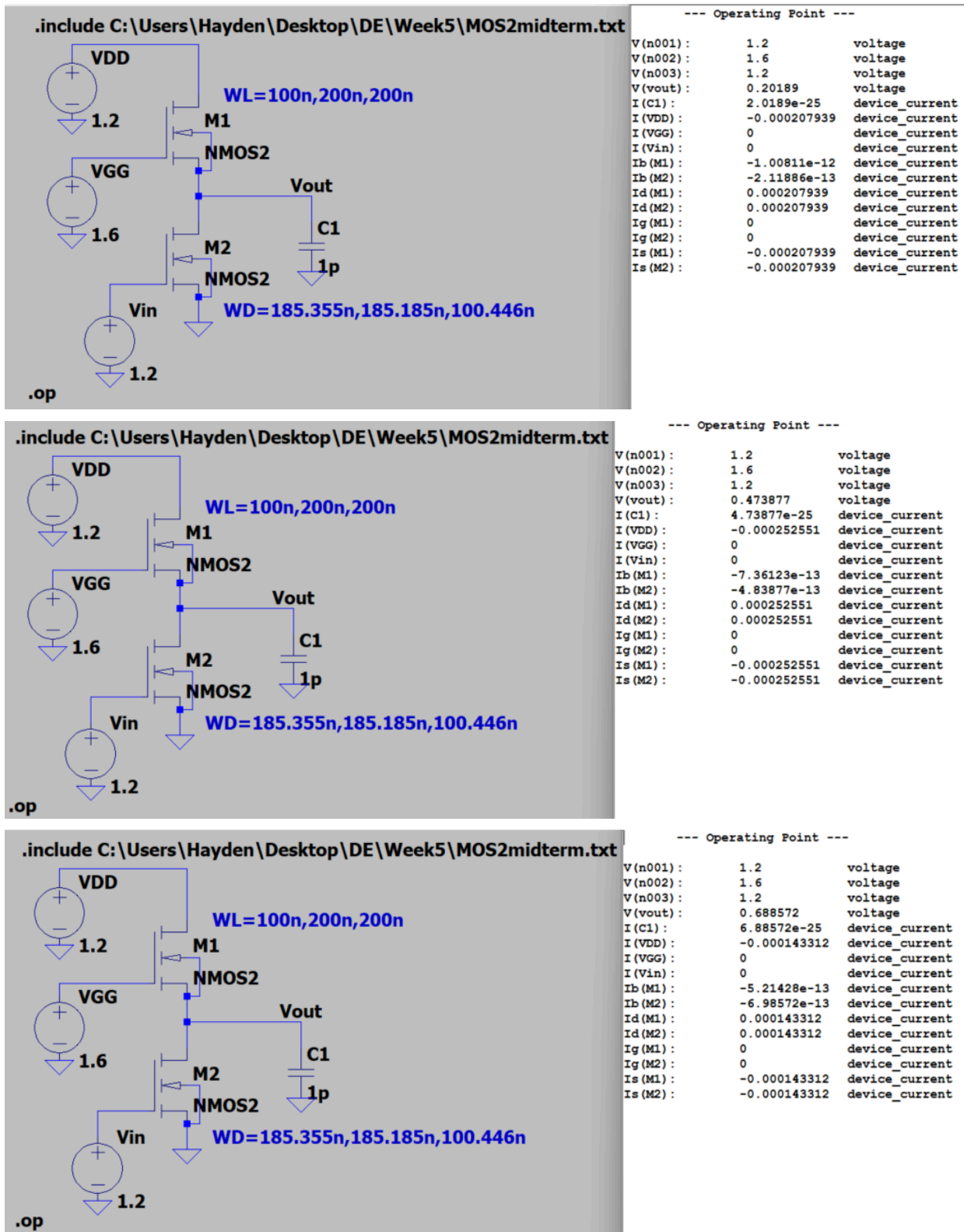
$$I_D = 1.3824 \times 10^{-6} W_D (1 + 0.2 * 0.6) = I_L = 2.16 \times 10^{-6} * W_L ((1.2 - 0.6)^2)$$

$$1.3824 \times 10^{-6} W_D (1.12) = 2.16 \times 10^{-6} * W_L (0.36)$$

$$W_D/W_L = (2.16 \times 10^{-6} * 0.36) / (1.3824 \times 10^{-6} * 1.12)$$

$$W_D/W_L = 0.50223$$

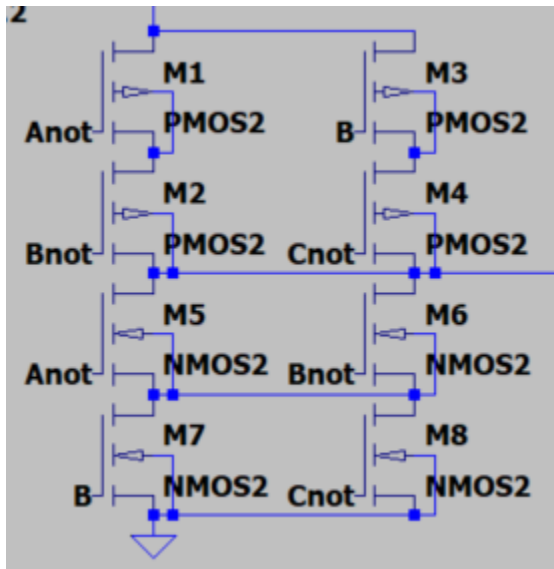
b)



All my LTspice Vouts are approximately 0.1V high, this is likely because I approximated with long channel equations

P3:

a)



$t_p \sim 0.69 R_{eq} C_L$

$R_{eq} = t_p / (0.69 \cdot C_L) = 50 \times 10^{-12} / (0.69 \cdot 75 \times 10^{-15}) = 966 \text{ ohm}$

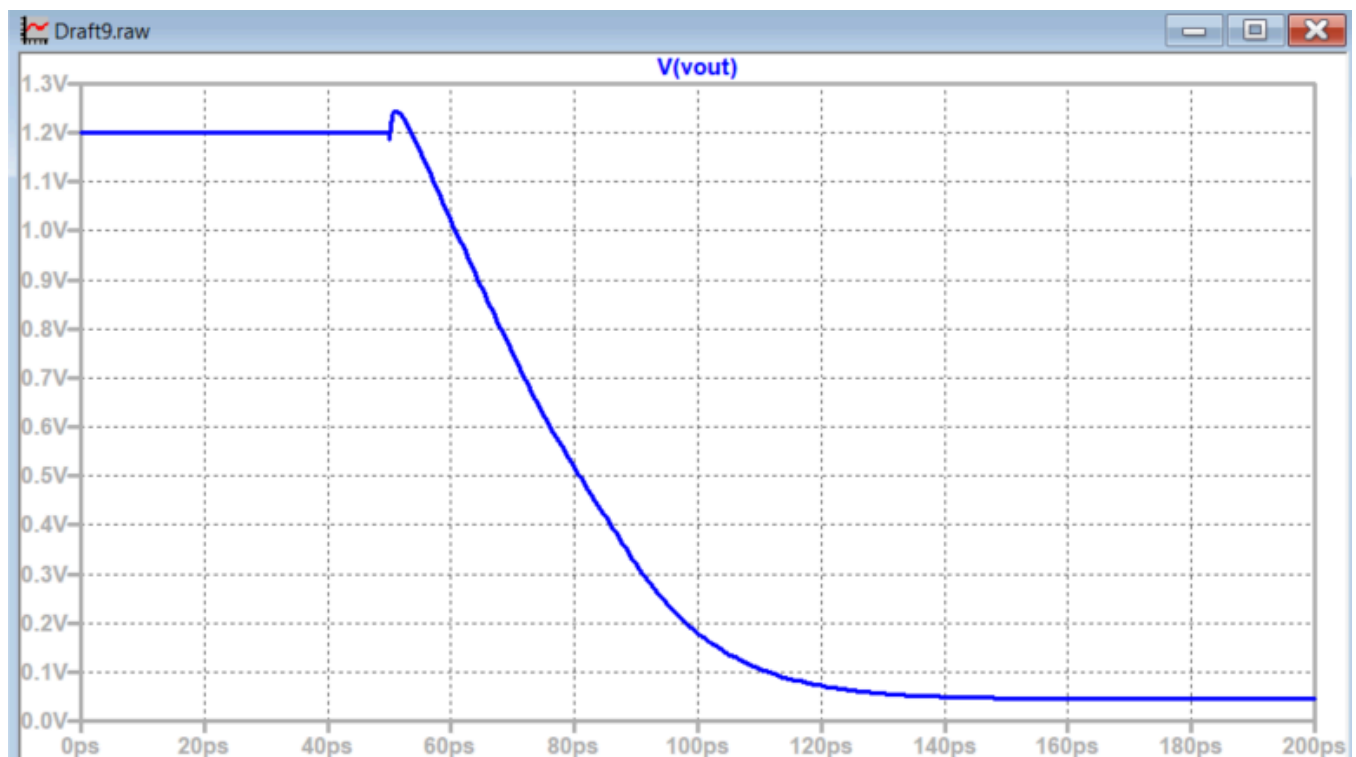
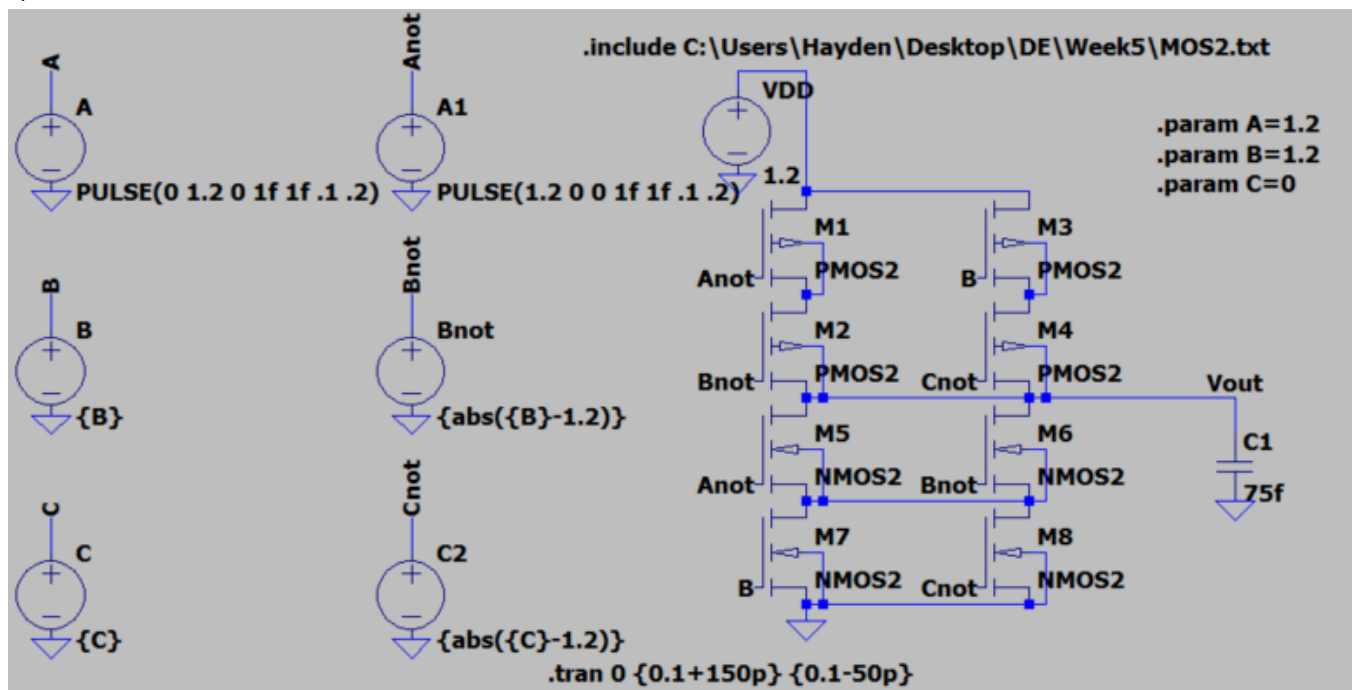
worst case for both is two in series, need half resistance each $966/2=483$

$483 = R_{on} \sim 1 / (\mu \cdot C_{ox} \cdot W/L \cdot (V_{DD} - V_T)) = 1 / (270 \cdot 1.6 \times 10^{-6} \cdot W / 100n \cdot (1.2/2 - 0.4))$

$W_n = 1 / (270 \cdot 1.6 \times 10^{-6} \cdot 483 / 100n \cdot (1.2/2 - 0.4)) = 2396.289 \text{ nm}$

$W_p = 270/70 \cdot W_n = 3.857 \cdot W_n = 9242.485 \text{ nm}$

b)



Seems to work fine across all possible state changes, crossing the threshold voltage in less than 50ps